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Review of cooling techniques using phase change materials for enhancing efficiency of photovoltaic power systems

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ABSTRACT

A significant fraction of the solar radiation falling on photovoltaic (PV) panels gets converted into heat thus reducing the efficiency of photovoltaic power plants. Generally the performance of PV panels falls by 0.5% for every degree rise in temperature, depending on the type of solar cells used. The temperature regulation of PV power systems therefore becomes important especially for areas experiencing consistently high temperatures to improve PV efficiency. The objective of the study is to review literature on photovoltaic cooling techniques using phase change materials (PCM) including PV-thermal systems and building integrated photovoltaic systems. The main aim is to identify important research areas to ensure reliable performance and commercially viability of the technology. Various parameters are required to be evaluated and optimized for the geographical location of interest to obtain the best output. Increase in electrical efficiency as high as 5% is observed with PV-PCM integrated systems. Study suggests that inorganic PCMs have a good potential for PV cooling. PCM based PV systems will be cost effective only for areas with high insolation year round with less inter-seasonal climatic variations. However, poor thermal conductivity and large undercooling are major problems associated with PCM. There are only few studies on performance testing, economic benefit identification, increase in PV panel lifetime and LCA analysis for testing the suitability of PCM as coolant for PV panels. In this study the research gaps are identified for follow up work. Market review suggests that PCM based PV cooling technology is not yet commercialized due to its inefficiency primarily because of technological challenges, high system costs and non-availability of reliable operating designs. The study suggests that PCM is still not an economic cooling technique for PV systems and further research is required.

1. Introduction

Current energy demands are met mostly by fossil fuels and non-renewable sources which are depleting at a fast rate besides being greenhouse gas emitters Therefore solutions based on renewable energy sources are being used to meet the current energy challenges. Solar energy is one of such clean and inexhaustible renewable energy resources with no carbon dioxide emissions and zero waste generation. Research efforts are being made over the years to effectively utilize the vast solar resource using photovoltaic (PV) panels but the efficiency is still low as a major part of solar energy is converted into heat energy. This not only reduces the electricity generation efficiency but also affects the lifetime of PV panels. It is found that PV performance falls with the increase in module temperature. The efficiency of crystalline silicon solar cells fall by 0.5% for every 1 °C rise in solar cell temperature and this decrease in efficiency varies with the type of cell [1]. The temperature of solar cell is a function of various parameters, most significant being the insolation received, wind speed over the surface of the PV panel, direction of the flow of the wind and ambient temperature. The temperature of the front and the back surfaces of the PV panel vary with the degree of orientation due to convection, conduction and radiation effects.

As the solar radiation falling on the PV panel, wind speed and ambient temperatures, are not under human control, research is focused to develop new materials and ways to reduce the solar cell temperature without adding much to the system cost. One technique is cooling by sprinkling water at the front or circulating at back of the PV panels. This method however consumes a lot of pump power. Another technique is the use of ventilated PV panels which is a current research focus. But this technique suffers from poor heat transfer rates and insufficient cooling. Further it become ineffective if the temperature rises too high. Also the heat removed cannot be very effectively stored or reused which could otherwise have helped in reducing the system cost as well as raising the overall efficiency of the system. Water and air

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based cooling provide immediate use of the heat stored so the flexibility of extending the time of stored energy use is limited.

Phase change materials (PCM) offer an attractive solution since they have several times more heat capacity than water and air based systems. Further the heat stored can be used during night without significant heat losses. The heat stored can then be used for air conditioning or water heating which raises the overall efficiency of the system and reduces its effective cost.

This work focuses on review of PV cooling literature using PCM to identify suitable materials, research gaps and future research areas.

The paper is organized into four sections: Section 2 provides overview of conventional cooling techniques followed by detailed literature review on PCM cooling for PV in Section 3. The identified research gaps and follow up research areas are presented in Section 4 followed by conclusions in Section 5.

2. Overview of conventional PV panel cooling techniques

2.1. Air cooling

2.1.1. Natural ventilation

This technique uses ambient air flow to reduce PV panel temperature by natural ventilation. Heat transfer can further be enhanced by increasing the heat transfer area using fins and extended surface designs. The drawbacks of this system are the fluctuations in PV cell temperature and very high panel temperatures during peak insolation [2]. Yun et al. [3] studied a ventilated PV façade and found that peak temperature of PV panel was 55.5 °C in comparison to 76.7 °C for the panel without ventilation.

2.1.2. Forced ventilation

This technique of heat removal is based on forced airflow at the front and back of the PV panels. It however consumes significant amount of fan power. Krauter et al. [4] found that active ventilation with forced convection can increase electrical output by 8%.

2.2. Water based cooling

2.2.1. Passive cooling

It is based on PV panel cooling using water without the use of pumps. It is found effective for PV cooling provided good thermal contact between the PV and the collector system is ensured. Further the excess heat stored in water should be continuously removed or utilized elsewhere.

2.2.2. Active cooling

Cooling performance is improved as the water flow velocity is increased. However, increase in volumetric flow per unit time means increase in power consumption [2]. Krauter [5] found that water flow can increase the electricity generation efficiency of the panel by 8–9% along with reducing the reflection losses.

2.3. Liquid immersion cooling

The technique involves removal of heat from the PV panels by immersing them in a dielectric liquid in an elongated tube. The liquid of such a refractive index is chosen so that it can concentrate the solar radiation onto the PV cells. This technique can maintain panel temperature in range 30–45 °C [2].

2.4. Thermoelectric cooling

Thermoelectric cooling is based on converting additional heat generated by the panel into electricity based on Peltier effect. This technique can increase the electricity generation efficiency of PV by 8–23% [6].

2.5. Other cooling techniques

Several studies are found in literature to cool PV using nano-fluid containing suspended metal particles. Sardarbaddi et al. [7] tested silica/water nano-fluid for PV cooling and found that increase in thermal efficiency up to 12.8% is possible. Yun et al. [8] studied the influence of MgO/water nano-fluid thickness on cooling performance for different concentration of MgO in water. Studies on magnetic nano-fluid and their performance under varying external magnetic fields are also found in literature. Gadhiri et al. [9] found that 39% improvement in cooling performance using ferro-fluid can be obtained compared to distilled water, with overall efficiency of the system improving by 76%. The presence of alternating magnetic field further enhances the thermal performance. Y G lee [10] tested the floating type PV array system using pultruded fiber (a fiber drawn through a hot die and coated with resins) reinforced polymeric plastic and found that system could be cost effective.

2.6. Cooling using PCM

The high latent heat capacity of PCM is utilized to maintain the PV panel at a fairly constant temperature. The heat stored can later be used for space heating, water heating and other purposes. However, the initial investment with these systems is quite high. A comparison of various PV panel cooling techniques discussed above and their suitability is summarized in Table 1.

It is clear that the natural ventilation based cooling is the most ineffective method while forced water and air cooling techniques are being used around the world for PV panel cooling. PCM has an added advantage of its ability to delay the temperature rise of panel without any electricity consumption. Further, the heat stored can be reused which further enhances the system efficiency. Studies on liquid immersion cooling are lacking though it also appears promising. For thermoelectric (TE) cooling, the focus is more on increasing the electrical energy output of the panel. Lower the panel temperature, lower is the TE efficiency. It is also to be noted that the temperature range mentioned for different techniques in Table 1 are broad ranges as the results are a function of different parameters including insolation received, ambient temperature, wind speed, the orientation of the panel etc. Above figures should only be used as an estimation of the maximum obtainable performance from these techniques as per literature. For comparing the real benefits of the different techniques, it is necessary to study a specific system under same conditions using different techniques.

3. Literature review of cooling techniques using PCM

This section reviews updated literature and presents highlights of important studies. The aim is to identify the research gaps and follow up research areas which can improve the current status of progress of the technology.

Electricity generation using photovoltaic are promising techniques that can meet the current energy challenges but the problem of overheating of the panels affect the energy conversion efficiency as well as the life of the module. Increased temperature raises the charge carrier concentration which raises the short circuit current due to reduced band gap but the open circuit voltage drops. This occurs due to increased recombination rates thereby reduces the overall power output from the cell. Thermal regulation of the PV panels is therefore important [11].

Different methodologies are found in literature on PV cooling where most of the works are focused on passive cooling systems using air flow and water circulation. These systems however could not maintain the cell temperature to an appropriate level. Active systems affect the system economics remarkably, primarily due to the large pumping costs involved in circulating air and water for effective PV cooling. The

Table 1
Comparative suitability of various PV cooling technique studies.

Sr. No.	Cooling techniques	Advantages	Limitations	Achieved PV panel temperature range & Reference
1	Natural ventilation	• Zero or negligible cost input • No electricity requirement • No maintenance	• Low conductivity and heat transfer rate • High PV panel temperature fluctuations • Very high peak insolation temperatures • Depends on wind direction and ambient air temperature • Heat removed can't be reused effectively	50–70 °C [2]
2	Forced Ventilation	• High heat transfer compared to natural ventilation • Do not depend on ambient conditions	• High installation cost • Consumes large amount of fan power • System is noisy • Heat removed isn't effective for later use. It is best utilized instantly	20–30 °C [2]
3	Liquid immersion cooling	• Larger heat dissipation • Better electrical performance • Less reflection losses	• Ionization of water is a problem • Liquid ingress due to leakage • Salt deposition is an issue • Corrosion may occur so appropriate care is must • Containment design and its weight • Dust deposition on ducts	30–40 °C [2]
4	Thermoelectric cooling	• Low initial investment • Low maintenance cost • No noise issues	• Cooling performance is a function of ambient air conditions [6] • Electricity consumption • Very effective heat removal is required to ensure reliable cooling	Not given; here the aim is to enhance electrical efficiency and not the cooling of the PV panels
5	Active water cooling	• Very effective in maintaining PV temperature • Better heat transfer rates compared to ventilation systems	• Installation cost is high • Regular maintenance cost • High pumping power requirement to maintain panel temperature • System is noisy • Instant energy use/heat removed cannot be effectively stored for use at later hours • Corrosion needs to be taken care of.	Upto 22 °C [5]
6	PCM	• No maintenance and electricity costs • Can maintain PV panel at constant temperature for long time • Delays panel temperature rise • Heat stored can be effectively reused • No noise issues	• Large initial investment • Cyclic stability of the material is a concern • Containment is a concern • Poor thermal conductivity	• 25–30 °C [20]

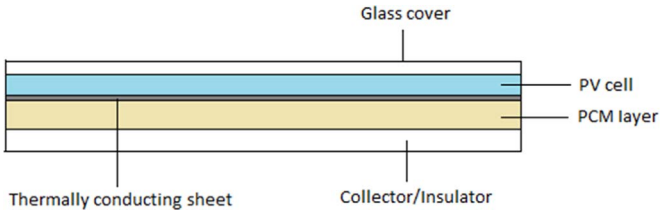


Fig. 1. Integration of PCM layer on backside of PV module for cooling.

focus of this review is on exploring the potential of PCM for PV cooling. Fig. 1 shows integration of PCM layer on the back side of the PV panel.

Literature reveals that PCM based PV systems when integrated with air conditioning or water heating systems could decrease the overall system cost while making it more effective. An example of the working system utilizing PCM technology is a Japanese house which uses air to cool PV panels. A grid connected 4.2 kWp system charges the PCM installed in the building roof and ceilings which reduce the building heating and cooling loads [12]. The highlights of the various reviews on cooling of PV panels are summarized in Table 2.

Ventilated air based cooling is found ineffective during peak hours. Water cooling has been found to outperform air based cooling. Other cooling techniques are at the primary stage of research. Studies in literature on identifying the suitable PCM's for PV cooling and their impact on the thermal and electrical efficiency are limited. Most of the studies are simulation /modeling and laboratory based. Some studies related to testing PCM performance and hybrid PV systems which include building integrated photovoltaic and PV thermal (PV-T) systems are carried out under field conditions in different countries. The limitation being the short time duration for which the studies are performed and the results are drawn based on the performance of PCM observed during that time. However, PCM does not result in positive performance every time which puts an important limitation on the reliability of the results. Not many studies are directed efforts towards designing systems where PCM waste heat could be re-used. The highlights of some important studies under field, laboratory and simulation modeling are summarized in the Table 3 with the aim to identify the potential PCMs suitable for cooling.

It is clear from the Table 3, that low melting PCM (LMPCM) couldn't maintain PV panel temperatures during the entire thermal regulation period. Though the increase in efficiency at the peak insolation hours for LMPCM is the highest, the overall increase in efficiency would be negative. However, a comparatively high melting PCM could maintain the PV panel at a constant temperature for the entire duration. CP was found to be a promising PCM among organic and $CaCl_2.6H_20$ among inorganic PCM's. It is also to be observed that the increase in efficiency with PCM is highest for area with consistently high solar radiation year round. Thickness and PCM thermal conductivity are other parameters which are found to influence the panel efficiency markedly.

C.J Ho et al. [29] tested the cooling performance of the PV with

Table 2
Highlights of PV cooling using various techniques.

Reference	Highlights
[1]	• Reviewed conventional cooling techniques of PV primarily water and air based cooling of PV • Most studies are performed for low PV ratings which create scale-up difficulties • Studies on degradation of PV panels with temperature are limited • PV cooling with micro channel, thermoelectric modules and automotive radiator are in developing stage and need further research
[2]	• Discussed cooling of PV module using natural/ forced, hydraulic, heat pipe and PCM cooling. • Large fluctuations in temperature of the PV cell are found in ventilated façade system • Surface area should be increased as the concentration ratio increase to maintain the cell temperature • Utilization of the waste heat from the PCM based system is not well studied
[13]	• Presents review literature on PV-PCM systems; • PV Performance is found to be affected by low heat transfer and large maintenance cost. Apart from this it involves large capital investment while its service life is low • Need to study benefits from implementation in real systems to access its market potential • Systems are monitored on laboratory scale; does not provide information about its suitability under real environmental conditions
[14]	• Reviewed cooling of the PV/T systems through single phase fluids • Water based cooling is more effective than air based cooling • Most of the cooling techniques are able to achieve efficiency > 10% • Direct comparison of various techniques is difficult due to irregularity of the conditions in which the performance is tested • Systems should be modeled based on non-dimensional number.
[15]	• Reviewed the cooling techniques for PV panels including impingement jet, micro-channel cooling, PCM, heat sink and heat pipe cooling • PCM suffers from toxicity, corrosiveness and inflammability. Their disposal after use is also a problem • Micro-channel heat sinks can achieve uniform low temperatures for PV panels • Hybrid micro-channel jet impingement technique achieves the minimum cell temperature. • Cooled PV panels were found to have lower energy payback time and greenhouse gas payback time

microencapsulated PCM and found that PCM with 30 °C melting temperature performed better than the one melting at 28 °C. They concluded that PCM couldn't completely solidify at night after a certain thickness thereby affecting its cooling performance. Japs et al. [30] found that the energy generated by the PCM-PV is higher than reference PV panel without PCM for 5 out of 25 days while with PCM+ graphite-PV, it was lower for all the 25 test dates studied. They found that the results of PCM used with PV are only positive in the forenoon at peak temperature hours while the daily average energy and economic yields were negative for the whole test duration. Further they [31] found that PCM equipped with graphite performed better than that without it as graphite enhances the conductivity of the PCM which causes rapid heat dissipation resulting in considerably improved performance.

Smith et al. [32] did global analysis of the PV panels using PCM and found that PCM is beneficial for high insolation areas with low inter seasonal climatic variations. An improvement in energy output of over 6% is seen on the Western coast of Mexico and improvements of over 5% are seen in many regions. For none of the regions studied, financial benefits were able to overcome the material costs assuming an average lifetime of 25 years of PV/PCM systems. The greatest improvements in performance were found for Africa, South Asia, Australia and South and Central America as these areas receive high levels of irradiance and often experience high ambient temperatures year-round.

The other works in literature on PV-PCM based thermal /building integrated / thermoelectric systems and important findings are summarized in Table 4. These include PV/T systems, BIPV systems, office buildings, study of various parameters affecting the PV-PCM performance and studies on modeling PV-PCM behavior.

Studies in Table 4 reflect that a considerable improvement in performance is observed when PCM is combined with the heat sinks. The number of fins, spacing and length are found to influence the conduction and convection behavior in PCM thereby influencing the melting time and the temperature regulation performance. Increase in electrical efficiency as high as 15% has been observed with this hybrid system. Increase in PV performance decreases the thermal performance of the system so a tradeoff must be made to get the maximum system efficiency. Modeled systems are not able to predict the solidification behavior properly and considerable discrepancies are observed.

4. Results and discussion: Identified Research gaps and follow-up research areas

The review suggests that literature is more focused on potential of organic PCM's as a cooling agent. The studies are found on testing the performance of integrated designs with PCM, PV-thermal and BIPV under laboratory and field conditions and modeling/simulation studies validated through experimental testing. The studies on determining the possible thermal benefits using PCM, resultant effects on its lifetime and year round increase in power generation efficiency are however limited. Based on the study of literature the status of research, identified research gaps along with follow-up research areas are presented and discussed in this section.

4.1. Criteria for selecting appropriate PCM

Most of the research highlights the impact of PCM on the PV performance for specific periods when the conditions are suitable for PCM i.e. high insolation and average ambient temperature. Performance under real conditions is also not well studied. The PV-PCM system essentially requires higher thermal conductivity and high heat capacity PCM. However, the criteria for selecting the PCM melting temperature (m.p.) are not well defined in literature. Lower melting PCMs (around 25 °C) are able to maintain temperature of PV panels in the desired range for very short intervals and become ineffective after peak insolation hours. Increasing thickness of the lower melting PCM create problems of incomplete melting. On the other hand, high melting PCM (> 30 °C) cannot maintain PV at the desired temperature of 25 °C although it can be maintained below a particular temperature for the entire duration. This essentially means that high melting PCM could maintain the panel at slightly higher but uniform temperature preventing creation of hot spots while low melting PCM could maintain the panel at low temperature at which it gives maximum efficiency but only for a limited duration. This suggests that using multiple PCM with different melting points could be a better solution. A balance must therefore be made between the PCM melting temperatures, thickness and the obtained efficiency.

4.1.1. Follow up research

PCM should be selected based on the peak PV temperature

Table 3
Identified PCMs for PV cooling and research highlights.

Ref.	PCM studied	Findings/Highlights	Temperature reduction (w.r.t no PCM case)	Study conditions
Studies in outdoor conditions/field testing				
[16]	• Petroleum Jelly	• Studied PV cooling using PCM under Indonesian climatic conditions. • Efficiency improves up to 21.6% for on roof systems and 6% for standalone systems due to heat removal by PCM. • PCM-PV is found more beneficial for building integrated designs where heat transfer from the back surface of PV is restricted.	• Reduced by 2.7 °C (on roof case) • Reduced by 0.4 °C (on-standalone case)	Tested two 10 W panels on the rooftop in standalone and rooftop mode in Indonesia
[17]	• Copper sulphate heptahydrate	• Able to maintain the PV panel temperature at 40 °C for around 2 h. at 1000 W/m^2 insolation • A power increase up to 7.92% was observed.	• Reduced by 4 °C	Not clearly defined
[18]	• Capric-Palmiticacid eutectic • Calcium hexahydrate	• Electricity generated, heat stored and fall in temperature were far higher for salt hydrate than CP. • For climatic conditions of Pakistan, electrical efficiency of PV with CP was 13.9% while 14.5% with salt hydrates. Thermal efficiency was 17.2% and 20.3% respectively. • Financial benefits are almost twice of the cost incurred for climatic conditions of Pakistan • Not suitable for climatic conditions of Ireland • The cost saving from the use of the PCM depends on the quantity of PCM purchased, geographical location and the climatic conditions	• Reduced by 21 °C with salt hydrate and 17 °C with CP for Pakistan • Reduced by 10 °C and 7 °C with salt hydrate and CP respectively for Ireland	65 W polycrystalline EVA encapsulated PV panels Outdoor testing in Dublin and Vehari Dublin – 27Aug–13 Sept, 2013 Vehari – 30 Oct- 13 Nov, 2013
[19]	• MP- 25 °CHeat of fusion −184 kJ/kg	• Effect of PCM on vertical polycrystalline silicon solar cells was studied. • For PCM melting temperature of 25 °C and 70 mm thickness maximum increase in efficiency of 4.6% • Optimum thickness of the PCM is determined as function of orientation of PV panels (70 mm South and 50 mm East). • Highest output was measured when panels faced south, where the solar radiation is the highest; lowest output of 5299 W h/y was measured when it faced east, where the solar radiation is the lowest. • PCM perform best in high insolation moderate air temperature areas	• Reduced by 10 °C for 100 mm PCM for 800 W/m^2	• 10.9 Wp polycrystalline cells • Climatic conditions of Song-do, Incheon, South Korea • Tested from 17 May–17 June 2012
[20]	• Capric Palmitic acid (PCM1) • $CaCl_2.6H_20$ (PCM2)	• For climatic conditions of Dublin, PCM2 maintained lower temperature than PCM1 though its melt temperature is higher (29.4 °C compared to 22 °C) • Daily averaged power saving of 1% for PCM1 and 1.8% for PCM2 for Dublin • For climatic conditions of Vehari, PCM1 dropped PV temperature to a peak of 16 °C while 21 °C for PCM2 against PV reference temperature of 63 °C • Daily averaged power saving of 4.4% for PCM1 and 7.7% for PCM2 for Vehari	PCM2 reduced temperature by 10 °C compared to 7 °C with PCM 1 against reference PV temperature of 49 °C	65 W polycrystalline EVA insulated panels tested Outdoor testing in Dublin and Vehari Dublin–27Aug–13 Sept, 2013 Vehari–30 Oct- 13 Nov, 2013
[21]	• RT35	• PV-PCM system studied; PCM could maintain PV temperature at 42 °C for 4 h. against a peak PV temperature of 53 °C	Reduction by 10 °C for 6 h.	Malaysian Climatic conditions
Simulation/modeling/laboratory testing studies				
[22]	• Coconut oil • Palm oil	• Palm oil was found to be a better PCM for ambient temperature range 27 °C–30 °C • Power output and efficiency are found to be the highest for the PCM with the maximum thickness • For 102 mm thick PCM, output power increased by 23% while electrical efficiency increased by 2.1%	Reduced by 9.6 °C for 102 mm thick PCM	10 W mono-crystalline panels Panels placed directly in contact with the roof of PAU building in Indonesia No details on the experimental conditions, no of days of testing
[23]	• Calcium hexahydrate • RT20 • Capric Lauric Acid (CL) • Capric-Palmitic acid (CP)	• Calcium hexahydrate has high heat capacity, high thermal conductivity and no volumetric expansion but undercooling up to 18 °C causes solidification problems at night. • Paraffin RT20 was found to have low density, high volume expansion, high flammability (154 °C) and very low solidification temperatures. • Paraffins have high meting range and large volumetric expansion which	Not mentioned	Lab testing using DSC and Temperature History Method

(continued on next page)

Table 3 *(continued)*

Ref.	PCM studied	Findings/Highlights	Temperature reduction (w.r.t no PCM case)	Study conditions
		makes them unsuitable for PV temperature regulation ● CL and CP suffer from wide melting range, low density and large volume expansion but has suitable solidification temperature ideal for summer nights in hot areas. ● CP appears to be the most promising PCM for PV temperature regulation among fatty acids.		
[24]	● RT20 ● SP22, ● Capric–Lauric acid ● $CaCl_2$ ● Capric-Palmitic Acid	● Low melting, low conductivity PCM in high conductivity containers perform better in low Insolation condition for PV ● High melting PCM in lower conductivity containers for Intermediate Insolation (750 W/m^2) and high melting, highly conductive PCM in high conductivity containers for high Insolation (1000 W/m^2) is suitable for temperature regulation of PV. ● $CaCl_2.6H_20$ maintained temperature reduction for the longest duration (11 h) followed by CL (9.5 h). ● $CaCl_2$ could maintain 10 °C temperature reduction for 5 h at 1000 W/m^2 insolation with Aluminum container ● CP and $CaCl_2$ that maintained a maximum of 18 °C temperature reduction at PV front surface for 30 min for 1000 W/m^2 with Al container ● Low conductivity PCM perform better in low insolation condition with high conductivity containers ● Highest temperature reduction was observed with the highest conductivity container for all the insolation conditions.	● For 1000 W/m^2, the duration for which different PCM maintained PV temperature below 10 °C is ● RT20 (2.6hrs) ● CL & SP22 (3 h) ● CP (4.2 h) ● $CaCl_2$ (5 h)	● Polycrystalline silicon PV cell of dimension 10 cm×10 cm×0.05 cm packed between perspex sheets to simulate a PV module ● Test conducted using solar simulator GR262 for 500 W/m^2, 750 W/m^2, 1000 W/m^2
[25]	● RT40	● Thermal conductivity of PCM infused graphite was 16.6 W/mK while 0.25 W/m K without it ● Incorporation of graphite extended the melting range of the PCM considerably ● Increasing thermal conductivity beyond 16.6 W/mK did not show any marked improvement except for very high Insolation's or very large PCM thicknesses (> 100 mm) ● At extended insolation, heat sinks has dominating behavior over PCM, ● Cost of thermal regulation ($130/m^2) was 10% of the solar panel	● Reduction by 19 °C in peak temperature for PCM + graphite + heat sink ● Reduced by 5 °C for PCM +graphite ● Shift in temperature rise with PCM + graphite wr.t reference	● 30 mm thickness for mono crystalline silicon PV panel area of 0.052 m^2 ● Insolation supplied using two Halogen lamp, 500 W each
[26]	● RT20 ● RT25 ● RT28HC	● For operational time of 480 min, only RT20 melt completely while RT25 and RT28HC melted by 80% and 65% respectively ● Maximum of 5% enhancement in efficiency was seen	● Temperature difference of 2 °C for 400 min with RT25 and RT28HC	● Model verified using Huang et al. experimental data assuming Polycrystalline cell of 10 cm × 10 cm × 0.05 cm ● Ambient temperature is assumed 19 °C
[27]	● RT10HC ● RT18HC ● RT25HC ● RT35HC	● Effect on transition temperature of PCM on the performance of PV was studied ● Increase in efficiency was highest for RT18HC (0.25%) followed by RT25HC (0.23%) ● Low transition PCM in the range of external temperature fluctuation should be chosen	–	ESP-r software used to study the system
[28]	● RT27-RT21 ● RT31-RT27	● Use of two PCMs for regulating the PV temperature in BIPV was studied ● RT27-RT21 was found to give the highest temperature reduction but the regulation period was the shortest. ● RT31-RT27 could maintain the temperature below 30 °C for the whole test duration. ● Low transition PCM's dominates the behavior of the two PCM system. ● Position of the PCM, system mass and PCM thermal characteristics are found to affect system behavior.	● < 30 °C throughout for RT31-RT27 ● 25 °C for RT27-RT21 for first 30 min after which the temperature rise rapidly	Simulation study

Table 4
PCM use in PV based thermal /building integrated / thermoelectric systems.

Ref.No.	Important findings	Study conditions
a).Type of PCM based PV System tested outdoors/Field testing		
[33]	• Compared PV/T-PCM system with PV/T system. • PCM used: Capric-palmitic acid with melting point 22.4 °C • PV/T-PCM-system heat storage capacity was found to be become twice than the PV/T system. • Heat availability of system with PCM increased by two times than the system without it.	Tested under climatic conditions of Dublin, South Africa during Sept 9-! 4,2014
[34]	• A PV/T/PCM system was tested for climatic conditions of Ireland • Systems were investigated for three days • Thermal efficiency observed ranged between 20–25%, and heat storage efficiency increased by 3.9%	Tested for climatic conditions of Dublin, Ireland PV rating not specified
[35]	• Studied multi crystalline PV cell performance with PCM RT27 of melting temperature 27 °C and heat capacity of 184 kJ/kg • A temperature difference of 5 °C was observed with PCM system as compared to naturally cooled system; 1% increase in electrical efficiency is observed. This temperature difference rose to 15 °C for a 12 fin–PCM system • CTEG-PCMTS (Concentrated thermoelectric generator) system is able to generate 8.47 W of thermoelectricity at 130 suns using 5 kg of paraffin based PCM.	Tested for the climatic conditions of Melbourne, Australia
[36]	• A BIPV-PCM system installed in office building was studied • Electrical efficiency of 10% and thermal efficiency of around 12% was found.	Tested for climatic conditions of Lisbon, Portugal Prototype dimensions 0.73×1.75 m Inner layer of PCM plasterboard Outer layer of PV polycrystalline modules with peak power 120 Wp
b).Simulation/Modeling/Laboratory testing based studies		
[37]	• Studied environmental benefits of incorporating PCM with PV • Studied influence of Glauber's salt and $Na_2HPO_4{\cdot}12H_2O$ on BIPV system for the climatic conditions of Zaragoza, Spain • Best performance was obtained with mixture of the two inorganic PCM's. • PCM melt temperature, thickness, wind speed are found to be critical parameters affecting system behavior. • These critical parameters should be selected as per the geographical location to obtain the right performance • A reduction in CO_2 emissions as high as 62 kg/m^2 in 25 years is possible with 1 °C temperature reduction and uniform temperature of the PV cells	Simulation study
[38]	• Studied performance of PV/T system in presence of MPCM–28 slurry (Micro-encapsulated PCM in water) • Average diameter of the MPCM was 18.2 μm • Temperature rise of module was slow (32.4 °C at start to 34.6 °C at finish line) in presence of MPCM • Increase in Reynolds from 1508 to 3496 led to increase in electrical output of the PV/T module from 103 W to 108.6 W. • Net efficiency of module increased from 77.8% to 80.8% when the MPCM's wt% increase from 0% to 10% • At solar radiation of 600 W/m^2, Re of 3000, MPCM conc. of 10%, the electrical efficiency was found to be 14.1% quite higher than the normal efficiency values of 10–12%	System was tested under lab conditions PV/T module area 1.28 m^2 Tested under two solar simulator
[39]	• Simulated the performance of PV system in presence of PCM and water. • On increasing the PCM thickness from 0 to 0.03 m, the PV output increased by 6.5% • Increasing the conductivity of the PCM by a factor of 10, PV output increased by 3% • PV/ST/PCM performance was 9% more compared to the PV only system • PV performance increase with decrease in ST performance	Simulation study
[40]	• Studied PV performance with RT25 and GR40 PCM's • GR40 could maintain panel temperature under 51 °C while RT25 could keep it under 32 °C for 150 min at 750 W/m^2 • Fin spacing of 8–12 mm are appropriate to maintain low temperature for longer times. • Straight fins of 36 mm leads to lowest surface temperature • Soft iron wire matrix type of fin design gave the most stable temperature with the PCM	Tested under solar simulator, compact MSR 1200HR, 1000 W metal halide lamp
[41]	• Studied a HP-PV/T system in presence of PCM for the climatic conditions of Beirut. • Optimized the system to determine the PV cell area, PCM melt temperature, no. of PCM spheres for given electrical and thermal energy requirements. • Increase in efficiency at peak hours as high as 8.7% is observed with PCM's • Heat pipe cooling increased the yearly power output by 4.47% • Change in PCM melt temperature has more influence on thermal performance than the number of PCM spheres • Decrease in melting temperature of the PCM increase the electrical efficiency while reducing the thermal performance	Modeling
[42]	• Heat and mass transfer in the solar panel modeled in the presence of PCM • Increasing PCM width is more effective than increase in height for cooling • Cooling fins increase conductivity but boost phase transition too. • PCM can maintain panel temperature under 40 °C for 80 min in insolation of 1000 W/m^2	Simulation study
[43]	• A Building Integrated Concentrated Photovoltaic system was studied in presence of PCM RT 40 • Use of PCM dispersed the heat uniformly preventing creation of hot spots • 13.7% improvement in electrical conversion efficiency was obtained compared to non-PCM case • A maximum 5.2 °C reduction in temperature with the T_{max} at center being 58.8 °C and T_{min} at center being 26 °C	Tested under Wacom solar simulator with solar intensity of 1000 W/m^2
[44]	• Three PCM's Waksol A, RT27 and RT35 were taken and effect of fin with these PCM's for PV thermal regulation is studied • Use of fins stabilized the temperature fluctuation in the system • More the no. of fins, lesser is the stratification in PCM temperature and lower is the convection • When fin spacing is reduced, conduction dominate the system and time of PCM melting is increased	Tested using GR262 solar simulator

(continued on next page)

Table 4 *(continued)*

Ref.No.	Important findings	Study conditions
[45]	• However very rapid rise in panel temperature occur after PCM is melted • Simulated the performance of the BIPV panel with MEPCM • For summer, use of MEPCM increased minimum efficiency by 0.13% while in winter by 0.33% • Aspect ratio (A_m) of MEPCM affect its impact on electrical efficiency, A_m −0.277 increased it by 0.09% while A_m=1 decreased it by 0.18% • MEPCM of melting temperature 34 °C performed better than 26 °C MEPCM, increasing minimum efficiency by 0.12% compared to 0.09% for 26 °C melting MEPCM	Modeling/simulation
[46]	• Developed 1D finite difference method to evaluate PV-PCM performance • Found that the governing parameter in PV-PCM performance is the solidification of the PCM at night.	Modeling
[47]	• Developed a theoretical model using finite element method for predicting panel temperature. • Predicted day temperatures well for the sunny and cloudy days. • Temperature calculated at night were far low than that measured.	Modeling

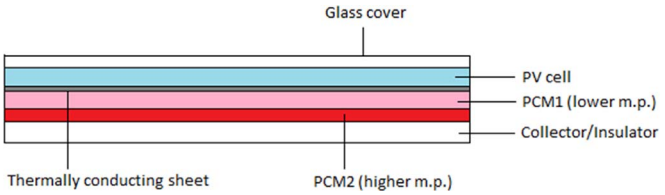


Fig. 2. Using multiple PCM layers for effective cooling.

observed (year round average) rather than the average ambient conditions. Multiple PCM's with different melting temperature need to be introduced in order to adjust to diurnal changes in ambient conditions. The solution could be using low melting PCM selected on the basis of average ambient temperature/required panel temperature placed directly in contact with the PV, then PCM with comparatively higher melting PCM, selected on the basis of peak panel temperature is placed to ensure high temperature difference for quick heat dissipation as shown in Fig. 2.

Low thermal resistance should be ensured between the two PCM layers so that the heat can be rapidly transferred between them. Using multiple PCM not only increase the heat capacity of the system but also serve to reduce the system weight and effective PCM thickness which causes problem of incomplete solidification and thermal stratification. However, the relative amount of low melting PCM with respect to high melting PCM to ensure complete melting during the thermal regulation period needs to be further conceptualized.

4.2. Salt hydrates for PCM cooling

Salt hydrates are potential candidates for cooling of PV panels in terms of their high conductivity, large heat storage capacity and lower volumetric expansion but suffer from large undercooling which creates solidification problems at night especially in summers when the thermal regulation need for PV panels is the highest. Organic PCM on the other hand suffers from poor conductivity, low heat capacity and high volumetric expansions but are free of sub cooling and super cooling problems. The performance improvements have been observed with organic PCM in spite of its low thermal conductivity however in none of the studied systems, economics advantage outperformed the invested costs. This serves as a strong motivation for working on salt hydrates which has far better conductivity while being several times cheaper than organic PCM.

4.2.1. Follow up research

Literature shows that potential of salt hydrates for cooling of the PV panels is not much studied due its inherent limitations i.e. very high sub-cooling. The only material tested for its PV cooling potential is $CaCl_2.6H_20$ which is able to reduce the panel temperature by as much as 21 °C [17] and maintain it at constant temperature for large duration in almost all the studies tested. If the challenges with their undercooling problems are met, economic advantages expected out of the use of salt based PCM are far higher as they are several times cheaper than their paraffin counterparts.

4.3. Solidification of the PCM at night

It has been observed that the solidification of the PCM at night results in performance problems. This is one of aspects ignored by most of research studies. The simulation models predicted PCM temperature well during the day but significant differences are found during the night. During solidification of the PCM heat transfer comprises of complex flow geometries and convection becomes important. Further the PCM detaches from the container wall adding resistance to heat transfer.

4.3.1. Follow up research

Solidification of the PCM at low temperatures and during night is a serious issue which requires focused research for the success of this technology. It is found that for high density materials, the top layers do not allow the lower layers to melt and take up most of the heat in the form of sensible heat storage. This prevents melting of the lower layers during the day times. This layer then adds to the thermal resistance and prevents the solidification of the top PCM layers during the night hours. Thus there is a need to properly model and optimize the size of PCM for a given system keeping these aspects in mind. There is also a need to develop new materials which are able to meet the criteria and cater to the system requirements. Further active systems must be conceptualized to ensure appropriate cycling in case the climate is not suitable to allow natural solidification. All these aspects needs to be further studied since the PCM solidification is an important constraint on its overall performance of the PCM based systems.

4.4. Flammability of the PCM

PCM's especially paraffins and fatty acids need to be carefully used since they are highly flammable. Since the back surface temperature of PV panels in high insolation areas can reach really high temperatures (> 80 °C) during peak insolation hours, it is necessary to ensure that PCM is safe to use up to this limit.

4.4.1. Follow up research

Not much literature is found on testing the flammability of PCM though it is an important consideration for its use. PV panels reach very high temperatures around 80 °C and even higher in hot climates and during peak insolation hours. Paraffins and fatty acids particularly need attention as they are highly flammable but are important candidates for PV cooling. Salt hydrates are however safe to use.

4.5. Status of PV-PCM system performance studies

Most of the PV-PCM system studies are based on modeling

validated through experiments under laboratory conditions which cannot account for the discrepancies found in performance under real climatic conditions like changing wind effects, non-uniform insolation, shading, ambient temperature changes etc. Very few performance studies on these systems under actual real conditions are carried out in Indonesia, Pakistan, Ireland, South Korea, Australia and Portugal. Also, for these systems too performance is only monitored for few days of the year.

4.5.1. Follow up research

More detailed research should be focused on long term monitoring of PV-PCM based systems under real climatic conditions. The PCM must be tested year round to analyze the fraction of time it enhances PV efficiency. It should then be compared with the fraction of time when efficiency falls in the presence of PCM. This occurs due to the heat accumulated in PCM which prevents the fall in PV temperature or causes PV overheating after peak hours when PCM completely melts. This comparison is essential to evaluate the real economic advantage of using PCM for cooling of PV systems.

4.6. Suitability, economic and environmental benefits of the use of PCM

Most of the studies identifies the suitability of the PCM based on the observed increase in performance for certain duration tested however the choice of PCM, orientation of the PV panels, PCM thickness, width and thermophysical parameters are a function of the location at which the system is to be installed. All the locations may not be suitable for PV cooling using PCM. The choice of PCM as a coolant has to be made based on the detailed analysis of the climatic conditions of the location under consideration.

4.6.1. Follow up research

Suitability of a PCM for cooling of PV panels is a strong function of geographical area and type of arrangement for best performance whether it has to be single PCM system, multiple PCM system, PCM with heat sinks etc. It is therefore necessary to test the PCM performance year round and perform detailed LCA analysis, expected enhancement in PV lifetime and efficiency due to cooling, expected payback time and study of fraction of the year when the influence of PCM is negative on the PV panel performance. These results should be carefully studied to find if the use of PCM would be an environmental friendly economic coolant choice.

4.7. Need for common test conditions for testing PCM performance

PV systems with different PCMs are studied in literature for their effect on PV power generation efficiency for different set of climatic conditions/ locations / months or days for the same region, making performance comparison difficult.

4.7.1. Follow up research

There is a need to develop some standard test conditions under which PV with PCM should be tested. The standard test conditions should be defined to test PCM performance for PV cooling for comparison of different PCMs. The conditions under which PV-PCM should be studied need further detailed analysis. This would ensure savings in time and energy in redesigning systems.

5. Conclusions

A comprehensive review of cooling techniques for photovoltaic power systems using phase change materials is presented to identify the status, research gaps and follow up research areas while also suggesting the plausible solutions. Based on the review, the main conclusions are as follows:

- PCM is not an economic and the most preferred cooling solution for PV panels as yet primarily due to its low thermal conductivity, unreliable solidification behavior, lack of enhancement in performance year round due to lack of flexibility in changing the parameters as per the need of the climate (like flow rate with water system) and high system costs.
- Suitability of the PCM for PV cooling depends on geographical location and the year round climatic conditions. PCM based cooling appears to be economical only for areas which experience high insolation with high ambient temperatures year round and experiencing less inter-seasonal climatic changes.
- Organic PCMs in spite of their low thermal conductivity are found to improve the electrical efficiency of the PV panels as high as 5% but for none of the systems, economic advantage using PCM outperformed the invested cost. This serves as a strong motivation to work on challenges presented by salt hydrates to be used for PV cooling.
- A study on potential of salt hydrates for PV cooling is limited in literature with the only material widely studied being $CaCl_2.6H_20$
- $CaCl_2.6H_20$, Capric-Palmitic acid (CP) and Capric Lauric acid (CL) appear to be suitable PCM's but $CaCl_2.6H_20$ suffers from cyclic instability while CL & CP suffers from poor conductivity and large melting range. These aspects needs to be further studied.
- PCM of melting temperature greater than 30 °C appear to be promising in maintaining constant PV panel temperature as compared to low melting PCM (< 25 °C). The former avoid formation of hot spots by keeping the panel at uniform temperature and also ensures proper solidification in hot summer nights. The latter however becomes ineffective after peak insolation hours and also face problems with complete solidification at night.
- PCM-PV cooling system are more economically suitable for building integrated designs in which heat dissipation from the back surface is difficult and where the possibilities of heat reuse are high.
- Suitability of the PCM as a coolant should be tested by more detailed performance studies year round under real outdoor conditions along with LCA and payback time analysis.
- Using heat sinks and conductivity enhancers with the PCM's are found to improve the overall efficiency of the PV panels several times.
- Focus should also be directed to developing new materials with better conductivity to meet the challenges in PV cooling technology.

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